$\Delta = \int_R$

$\boxed{T11}$

$H_0: \xi \sim P_0(x) = 1 \{(0,1)\}$

$H_1: \xi \sim P_1(x) = \frac{e^{1-x}}{e-1} \{(0,1)\}$

а) построить наиболее мощный критерий проверки

$n = 1$ α

$$l(x) = \frac{L_1(x)}{L_0(x)} = \frac{\frac{e^{1-x}}{e-1}}{1} = \frac{e^{1-x}}{e-1} \geq C_1$$

$G: x \leq A$

$$P(x \leq A \mid H_0) = \alpha \Rightarrow \int_0^A 1\,dx = \alpha$$

$$A = \alpha$$

$G: x \leq \alpha$

$$W = P(x \leq A \mid H_1) = \int_0^\alpha \frac{e^{1-x}}{e-1}\,dx = \frac{e}{e-1}\int_0^\alpha e^{-x}\,dx = \frac{e}{e-1}(-e^{-\alpha}+1)$$

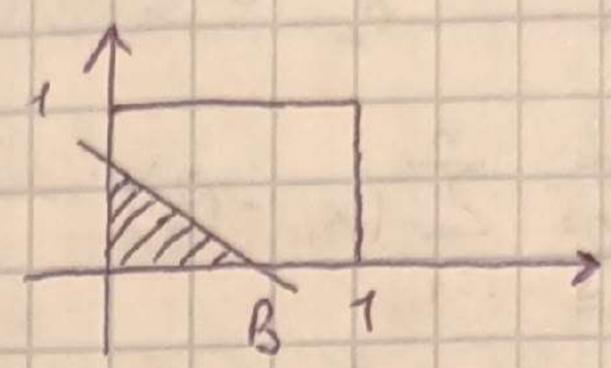
$\alpha_1 = \alpha$ $\alpha_2 = 1 - W$

в) построить наиболее мощный критерий проверки

$n=2 \qquad \alpha$

$$l(\vec{x}_n) = \frac{L_1(\vec{x}_n)}{L_0(\vec{x}_n)} = \frac{\frac{e^{1-x_1}}{e-1} \cdot \frac{e^{1-x_2}}{e-1}}{1 \cdot 1} = \frac{e^{2-x_1-x_2}}{(e-1)^2} \geq C_2$$

$$G: \ x_1 + x_2 \leq B$$



$$P(x_1 + x_2 \leq B \mid H_0) = \alpha$$

$$\iint\limits_{\substack{x_1+x_2 \leq B \\ 0 \leq x_1 \leq 1 \\ 0 \leq x_2 \leq 1}} 1 \cdot 1 \, dx_1 dx_2 = \alpha$$

(при малых $\alpha \leq \frac{1}{2}$) $\Rightarrow B \leq 1$

$$\frac{B^2}{2} = \alpha$$

$$B = \sqrt{2\alpha}$$

$$G: \ x_1 + x_2 \leq \sqrt{2\alpha}$$

$$W = P(x_1 + x_2 \leq B \mid H_1) = \iint\limits_{\substack{x_1+x_2 \leq B \\ x_1 \in [0,1] \\ x_2 \in [0,1]}} \frac{e^{1-x_1} \cdot e^{1-x_2}}{(e-1)^2} dx_1 dx_2 =$$

$$= \frac{e^2}{(e-1)^2} \int_0^B e^{-x_1} dx_1 \int_0^{B-x_1} e^{-x_2} dx_2 = \frac{e^2}{(e-1)^2} \int_0^B e^{-x_1} \left(1 - e^{-B+x_1}\right) dx_1 =$$

$$= \frac{e^2}{(e-1)^2} \left(1 - e^{-B} - Be^{-B}\right)$$

$\alpha_1 = \alpha \qquad \alpha_2 = 1 - W$

с) построить наиболее мощный асимптотич. критерий

$n \quad \alpha$

(исследовать состоятельность)

$$l(\vec{x}_n) = \frac{L_1(\vec{x}_n)}{L_0(\vec{x}_n)} = \frac{\prod_{i=1}^n P_1(x_i)}{\prod_{i=1}^n P_0(x_i)} = \prod_{i=1}^n \frac{P_1(x_i)}{P_0(x_i)} \geq C_3$$

$$\ln l = \sum_{i=1}^{n} \ln \frac{P_1(x_i)}{P_0(x_i)} \overset{\text{из } x_i}{\geq} \ln C_3$$

$$\sum_{i=1}^{n} \ln P_1(x_i) - \sum_{i=1}^{n} \ln P_0(x_i) = \sum_{i=1}^{n} \ln\left(\frac{e^{1-x_i}}{e-1}\right) - 0 \geq \ln C_3$$

$$\sum_{i=1}^{n} (1-x_i) - n\ln(e-1) \geq \ln C_3$$

$$\sum_{i=1}^{n} x_i \leq D$$

$$G: \quad \bar{x} \leq E$$

$$P(\bar{x} \leq E \mid H_0) = \alpha$$

Простейшая ЦПТ $\frac{\frac{1}{n}\sum_{i=1}^{n} \eta_i - M\eta_1}{\sqrt{D\eta_1}}\sqrt{n} \rightsquigarrow N(0,1)$

$H_0: \ M\xi = \frac{1}{2} \qquad D\xi = \frac{1}{12}$

$$P(\bar{x} \leq E \mid H_0) = P\left(\frac{\bar{x} - 1/2}{\sqrt{1/12}}\sqrt{n} \leq \frac{E - 1/2}{\sqrt{1/12}}\sqrt{n}\right) = \alpha$$

$$\frac{E - 1/2}{\sqrt{1/12}}\sqrt{n} = u_\alpha$$

$$E = \frac{1}{2} + \frac{u_\alpha}{\sqrt{12n}}$$

$$G: \ \bar{x} \leq \frac{1}{2} + \frac{u_\alpha}{\sqrt{12n}}$$

$$W = P(\bar{x} \leq E \mid H_1)$$

$$H_1: \ M\xi = \int_0^1 x \frac{e^{1-x}}{e-1}\,dx = \frac{e}{e-1}\left(-\int_0^1 x\,de^{-x}\right) = \frac{e}{e-1}\left(-xe^{-x}\Big|_0^1 + \int_0^1 e^{-x}dx\right) = \frac{e}{e-1}\left(-e^{-1} - e^{-1} + 1\right) = \frac{e}{e-1}\cdot\frac{-2+e}{e} = \frac{e-2}{e-1}$$

$$M\xi^2 = \int_0^1 x^2 \frac{e^{1-x}}{e-1}\,dx = \frac{e}{e-1}\left(-\int_0^1 x^2\,de^{-x}\right) = \frac{e}{e-1}\left(-x^2e^{-x}\Big|_0^1 + \int_0^1 2xe^{-x}dx\right) = \frac{e}{e-1}\left(-e^{-1} - 2xe^{-x}\Big|_0^1 + \int_0^1 2e^{-x}dx\right) =$$

$$= \frac{e}{e-1}\left(-e^{-1} - 2e^{-1} + 2 - 2e^{-1}\right) = \frac{e}{e-1}\cdot\frac{-5+2e}{e} = \frac{2e-5}{e-1}$$

$$D\xi = M\xi^2 - (M\xi)^2 = \frac{2e-5}{e-1} - \frac{(e-2)^2}{(e-1)^2} = \frac{2e^2-7e+5-e^2+4e-4}{(e-1)^2} =$$

$$= \frac{e^2-3e+1}{(e-1)^2}$$

$$P(\bar{x} \le E \mid H_1) = P\left(\frac{\bar{x} - \frac{e-2}{e-1}}{\sqrt{\frac{e^2-3e+1}{(e-1)^2}}}\sqrt{n} \le \frac{E - \frac{e-2}{e-1}}{\sqrt{\frac{e^2-3e+1}{(e-1)^2}}}\sqrt{n}\right) =$$

$$= \int_{-\infty}^{\gamma} \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\,dx, \quad \text{где} \quad \gamma = \frac{\frac{1}{2} + \frac{u_\alpha}{\sqrt{12n}} - \frac{e-2}{e-1}}{\sqrt{\frac{e^2-3e+1}{(e-1)^2}}}\sqrt{n}$$

$\alpha_1 = \alpha \qquad \alpha_2 = 1 - W$

Состоятельность критерия: $W \xrightarrow[n\to\infty]{} 1 \quad \forall$ распр. из H_1

$$\gamma = \frac{\frac{u_\alpha}{\sqrt{12}} + \left(\frac{1}{2} - \frac{e-2}{e-1}\right)\sqrt{n}}{\sqrt{\frac{e^2-3e+1}{(e-1)^2}}} \xrightarrow[n\to\infty]{} +\infty \Rightarrow W \xrightarrow[n\to\infty]{} 1$$

критерий состоятельный

d) построить оптимальный критерий

$n \qquad G: x_{min} < C \qquad \alpha$ (исслед. состоят)

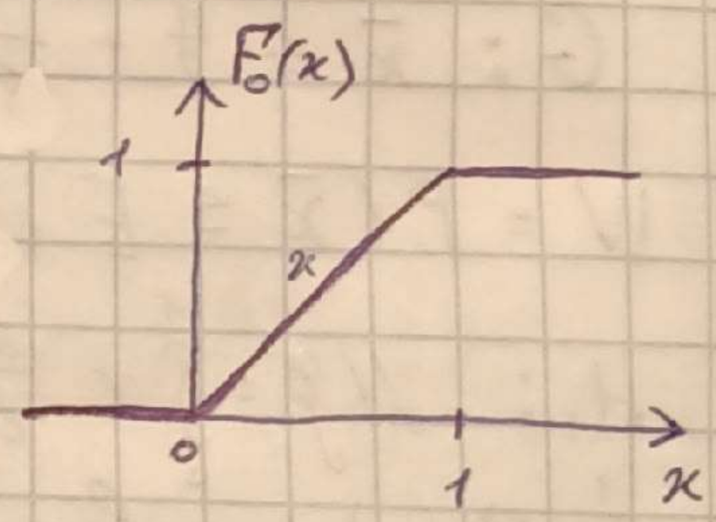
$\alpha_2 \to \min$

$\alpha_1 \le \alpha$

$$\alpha_1 = P(x_{min} < C \mid H_0) = 1 - (1 - F_0(C))^n \le \alpha$$

$$1 - (1 - C)^n \le \alpha$$

$$C \le 1 - \sqrt[n]{1-\alpha}$$



$$\alpha_2 = P(x_{min} \ge C \mid H_1) =$$

$$= 1 - P(x_{min} < C \mid H_1) = 1 - (1 - (1 - F_1(C))^n) = (1 - F_1(C))^n$$

$$F_1(x) = \int_{-\infty}^{x} \frac{e^{1-t}}{e-1}\,dt = \frac{e}{e-1}\left(-e^{-t}\right)\Big|_{-\infty}^{x} = \frac{e}{e-1}\left(1 - e^{-x}\right)$$

$$\alpha_2 = \left(1 - \frac{e}{e-1}\left(1 - e^{-C}\right)\right)^n \longrightarrow \min$$

$$f(C) = 1 - \frac{e}{e-1}\left(1 - e^{-C}\right) = \frac{e^{1-C} - 1}{e-1}$$

$$f'(C) = \frac{e}{e-1}\left(-e^{-C}\right) < 0 \quad \forall C$$

$$f'(c) \xrightarrow{\quad - \quad} c$$

$$f(c) \searrow 1-\sqrt[n]{1-\alpha}$$

$\Rightarrow \min \alpha_2 \Leftrightarrow \min f(c) \Rightarrow c = 1-\sqrt[n]{1-\alpha}$

$G: x_{min} < 1-\sqrt[n]{1-\alpha}$

$W = P(x_{min} < C \mid H_1) = 1-(1-F_1(C))^n =$

$= 1-\left(1-\frac{e}{e-1}(1-e^{-c})\right)^n$

$\sqrt[n]{1-\alpha} = e^{\frac{1}{n}\ln(1-\alpha)} = 1+\frac{1}{n}\ln(1-\alpha)+O\left(\frac{1}{n}\right)$

$e^{-(1-1-\frac{1}{n}\ln(1-\alpha)-O(\frac{1}{n}))} = e^{\frac{1}{n}\ln(1-\alpha)+O(\frac{1}{n})} = 1+\frac{1}{n}\ln(1-\alpha)+O\left(\frac{1}{n}\right)$

$W = 1-\left(1-\frac{e}{e-1}\left(1-1-\frac{1}{n}\ln(1-\alpha)-O\left(\frac{1}{n}\right)\right)\right)^n = 1-\left(1+\right.$

$\left.+\frac{1}{n}\frac{e\cdot\ln(1-\alpha)}{e-1}+O\left(\frac{1}{n}\right)\right)^n \xrightarrow[n\to\infty]{} 1-e^{\frac{e\cdot\ln(1-\alpha)}{e-1}} = 1-(1-\alpha)^{\frac{e}{e-1}} \neq 1$

не является состоятельн

$\alpha_1 = \alpha \qquad \alpha_2 = 1-W$